

Lab 5

Implement Amazon DynamoDB

AWS Technical Essentials • ~15 Minutes

🏷️ Resource Names for This Lab

Replace `<ID>` with your assigned student ID (for example `user7`).

Resource	Name
DynamoDB table	`Employees-<ID>`

> 💡 **Note:** table names are unique per Region per account, so every student needs their own. Item keys (`E001`, `E002`, `E003`) live inside your table and do not need a suffix.

🎯 Lab Overview

📋 Objectives

- Create a DynamoDB table with a partition key
- Add items with different attributes
- Scan and query the table
- Edit an item to demonstrate schema flexibility

📝 Lab Details

- **Duration:** ~15 minutes
- **Capacity mode:** On-demand
- **Region:** Same as previous labs

📊 Create the Employees-<ID> Table

1. Navigate to **DynamoDB** in the console
2. Click **Create table**
3. Table name: `Employees-<ID>`
4. Partition key: `id` (String)
5. Table settings: **Customize settings**
6. Capacity mode: **On-demand**
7. Leave all other settings as default
8. Click **Create table**

> ✅ **Expected:** Table status changes to **Active** within a few seconds.

📝 Add Employee Items

1. Click on the **Employees-<ID>** table, then **Explore table items**
2. Click **Create item**

Add these three items (click **Add new attribute** for each extra field):

id	name	department	location
E001	John Smith	Engineering	Seattle
E002	Jane Doe	Marketing	New York
E003	Bob Johnson	Sales	Austin

> 💡 **Tip:** For each item, click **Add new attribute** → **String** to add name, department, and location fields.

🔍 Scan and Query the Table

Scan (returns all items):

1. In **Explore table items**, select **Scan**
2. Click **Run** — all 3 items appear

Query (search by partition key):

1. Switch to **Query**
2. Partition key value: `E001`
3. Click **Run** — only John Smith's record appears

> ✅ **Expected:** Scan returns all 3 items. Query returns only the matching item.

🔄 Edit an Item — Schema Flexibility

1. Click on item **E003** (Bob Johnson)
2. Click **Add new attribute → String**
3. Add: `phone` = `512-555-0199`
4. Click **Save changes**
5. Run a **Scan** again — notice E003 has a `phone` attribute that other items do not have

> ✅ **Expected:** E003 now has 5 attributes while E001 and E002 still have 4. Each item can have different attributes.

🗑️ Clean Up

1. Go to **DynamoDB → Tables**
2. Select the **Employees-<ID>** table
3. Click **Delete**
4. Type `confirm` and click **Delete table**

> ✅ **Expected:** The Employees-<ID> table is deleted. DynamoDB deletes all items and indexes with the table.

🎉 Lab 5 Complete!

DynamoDB Table Created and Queried

🎯 Key Takeaways

🛠️ What You Did

- Created a DynamoDB table with a partition key
- Added items with varying attributes
- Used Scan and Query operations
- Demonstrated schema flexibility

📚 Key Concepts

- **Partition key** uniquely identifies items
- **Schema-flexible** — items can differ
- **Query** is efficient, **Scan** reads all
- **On-demand** = pay per request



Questions?

Ask your instructor before proceeding to the next module